

A 3D rendering of the text "DL for ENG" in a bold, gold, sans-serif font. The letters are floating on a deep blue ocean with visible ripples and small waves. The sky above is a mix of light blue and grey, with soft, white clouds. The overall scene has a cinematic, slightly surreal quality.

DL for ENG

## LECTURE 12.2

# Prediction of Power Grid Stability

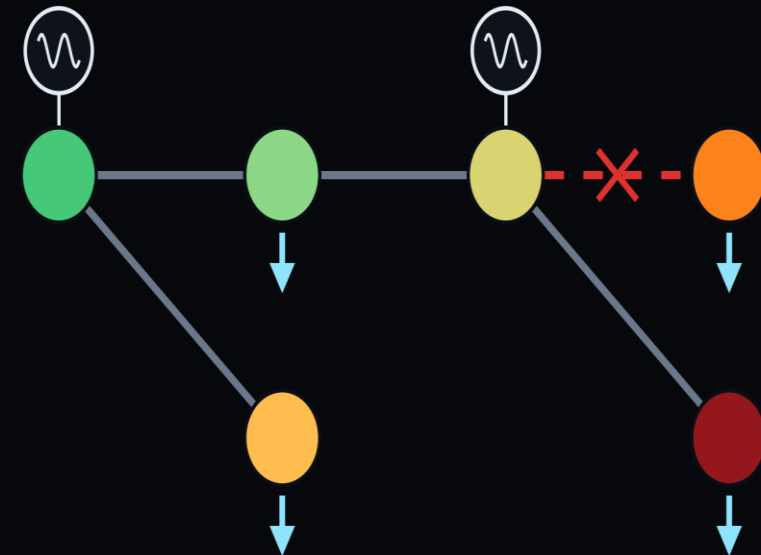
Estimation asked what the grid is doing. This asks where it is going — under a disturbance that is not in your training set, on a network that is not the one you trained on.

**Remus Teodorescu**

*With support from Research Assistant Noman Khan - nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk · Deep Learning for Engineering · 2026

*Transient stability · physics-informed GNN · N-1 screening · inference at the edge*



  
predicted stability margin after the fault    stable → critical

# What You Will Be Able To Do

## THE GOALS, IN PLAIN WORDS

- **distinguish** — the three questions people call stability, and say which one this is
- **explain** — why screening is combinatorial, and what a learned predictor is actually for
- **read** — the equal area criterion as an energy statement, and get a margin from it
- **state** — why the cases you must get right are the ones you have least of
- **represent** — a contingency as a change to the graph, not a change to a feature vector
- **argue** — why held-out topologies is the honest test, and held-out samples is not

## WHAT TODAY ASSUMES

|            |                                   |
|------------|-----------------------------------|
| from L12.1 | the swing equation, and the graph |
| from L5.1  | message passing, and equivariance |
| from L6.2  | compounding error in a rollout    |
| from L7.2  | why it needs two conditions       |

## ESTIMATION WAS INTERPOLATION

Extrapolation now; the residual survives.

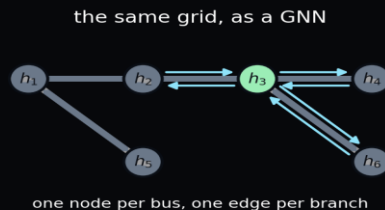
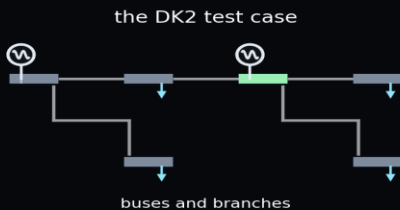
## AND THE LAST LECTURE

It closes with what the course has been teaching.

# Graph Networks, and Why Equivariance Matters Here

## WHAT YOU ALREADY HAVE

- the same message-passing network as L12.1, now predicting forward in time
- **equivariance is the property that is appropriate here** — permute the buses, the answer permutes
- weight sharing makes the update local, so it transfers to a grid it has not seen



## THE NETWORK, CONCRETELY

|                  |                                      |
|------------------|--------------------------------------|
| <b>inputs</b>    | pre-fault state, post-fault graph    |
| <b>hidden</b>    | message passing, shared weights      |
| <b>output</b>    | a stability margin from a trajectory |
| <b>from L7.1</b> | the swing residual in the loss       |

## THE CONTINGENCY IS AN INPUT

Only expressible if the graph is an input.

## AND THE RESIDUAL IS L7.1's

An ODE per machine, by autograd. New operator.

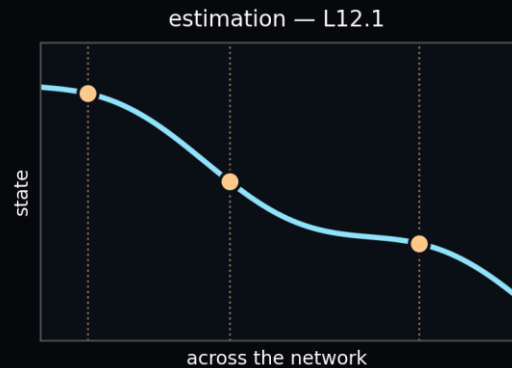
# From What Is, to What Will Be

## THE STATE YOU RECOVERED IS AN INITIAL CONDITION

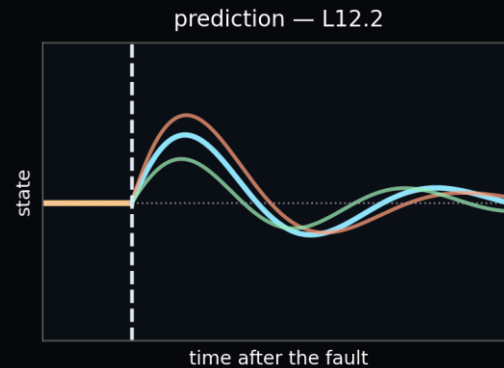
$$\mathbf{x}(t) = \{\delta_i(t), \omega_i(t)\}, \quad \mathbf{x}(0) = \hat{\mathbf{x}}_{L12.1}$$

$\delta_i$  rotor angle,  $\omega_i$  speed,  $\hat{\mathbf{x}}_{L12.1}$  the estimated state

- L12.1 produced a state consistent with the measurements and the physics



measurements pin it down, the physics fills the gaps



the data stops at  $t = 0$ , the physics carries the rest

## WHAT CHANGES

|                 |                                    |
|-----------------|------------------------------------|
| <b>question</b> | where is it going, not where is it |
| <b>data</b>     | stops at $t = 0$                   |
| <b>physics</b>  | carries the whole prediction       |
| <b>error</b>    | compounds instead of averaging out |
| <b>topology</b> | changes — that is the point        |

## WHY THE ESTIMATE MATTERS TWICE

A one per cent angle error at a instrumented bus is unremarkable in an estimate and can decide stable from unstable a second later. Carry L12.1's worst-bus error into this lecture, not its mean.

# Rotor-Angle, Frequency and Voltage Stability

## THREE QUESTIONS, THREE TIMESCALES

- **rotor angle** — do the machines stay in step after a fault? Seconds, governed by the swing equation
- **frequency** — does frequency stay within limits after a large imbalance? Seconds to minutes, inertia then reserves
- **voltage** — can the network hold its voltages after a disturbance? Minutes, reactive support and load recovery

## WHAT THIS LECTURE PREDICTS

Rotor-angle stability in the first seconds after a fault — the large-disturbance case the classification calls TRANSIENT — with the classical machine model.

Frequency enters only through RoCoF, which shares the same inertia term as the swing equation. RoCoF and nadir are what protection acts on.

## AND WHAT IT DOES NOT

Voltage stability needs load dynamics and reactive control the classical model does not contain. Say so in your report rather than letting a reader assume the prediction covers it.

# Given This State and That Fault, What Happens?

## A WELL-POSED QUESTION WITH AN EXPENSIVE ANSWER

$$\frac{2H_i}{\omega_s} \ddot{\delta}_i + D_i \dot{\delta}_i = P_{m,i} - P_{e,i}$$

$H_i$  inertia,  $D_i$  damping,  $\omega_s$  synchronous speed,  $P_{m,i} - P_{e,i}$  the power imbalance

- take the estimated state, apply a fault, clear it, integrate the swing equation
- the trouble is not formulating it
- the trouble is that an operator must ask it for every credible contingency, continuously
- everything in this lecture is a response to that cost

### THE INGREDIENTS

|                       |                                 |
|-----------------------|---------------------------------|
| <b>x(0)</b>           | the estimated state, from L12.1 |
| <b>the fault</b>      | where, what type                |
| <b>clearing time</b>  | when protection acts            |
| <b>post-fault net</b> | topology after the trip         |
| <b>the verdict</b>    | bounded, or not                 |

### NOTE THE FOURTH INGREDIENT

The network after the fault is not the network before it. A tripped line changes  $Y$  and the equilibrium the machines seek. Any method assuming a fixed topology has assumed away the problem.

# Time-Domain Simulation, and Why It Is Not Enough

## ALWAYS RIGHT, AND SOMETIMES TOO SLOW

- integrate the differential-algebraic system and you get the trajectory, to any accuracy you pay for
- the cost is per contingency and it does not amortise
- screening a large system against thousands of cases is not a real-time activity
- **so the industry runs a hierarchy** — cheap filters first, full simulation on survivors

### WHAT SIMULATION GIVES YOU

|                  |                                     |
|------------------|-------------------------------------|
| <b>accuracy</b>  | as much as the model deserves       |
| <b>detail</b>    | controls, protection, limits        |
| <b>trust</b>     | decades of validation               |
| <b>diagnosis</b> | the whole trajectory, not a verdict |

### WHAT IT COSTS

|                 |                                       |
|-----------------|---------------------------------------|
| <b>per case</b> | no reuse between contingencies        |
| <b>scaling</b>  | cases grow combinatorially            |
| <b>latency</b>  | too slow to sit inside a control loop |

# Why Screening Is a Combinatorial Problem

## MORE QUESTIONS THAN THE GRID HAS PARTS

- N-1 security means surviving the loss of any single element
- on a few hundred branches that is already hundreds of cases
- multiply by fault location, type and clearing time, and by a state that changes every few minutes
- **a learned predictor is not there to beat integration on accuracy** — it is there to answer many cases at once

## WHERE THE CASES COME FROM

|                        |                                      |
|------------------------|--------------------------------------|
| <b>single outages</b>  | branches, generators, transformers   |
| <b>fault type</b>      | three-phase, line-to-ground, ...     |
| <b>location</b>        | along the line, not just at the ends |
| <b>clearing</b>        | nominal, and delayed by backup       |
| <b>operating point</b> | and all of it again, next hour       |

## THE HONEST SCOPE OF THE EXERCISE

You will screen single-branch outages on a small network, which is a few dozen cases and runs in minutes. The combinatorial argument is why the method matters at real scale; do not claim you demonstrated it at real scale.

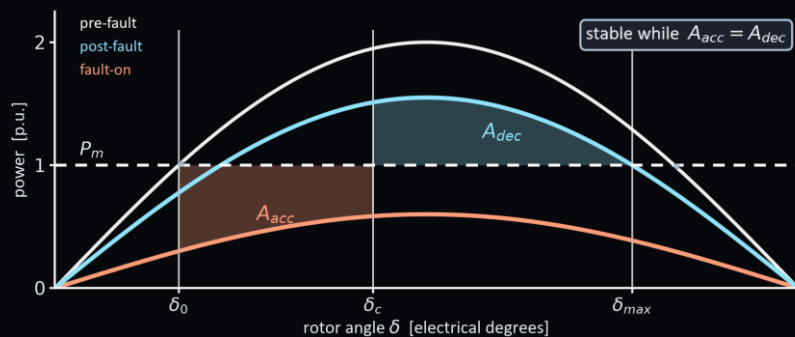
# The Equal Area Criterion

## WHY A MACHINE COMES BACK, OR DOES NOT

$$\int_{\delta_0}^{\delta_c} (P_m - P_e) d\delta = \int_{\delta_c}^{\delta_m} (P_e - P_m) d\delta$$

$\delta_0$  pre-fault,  $\delta_c$  clearing,  $\delta_m$  maximum angle,  $P_m$  mechanical,  $P_e$  electrical power

- during the fault the machine accelerates, because electrical power collapses
- read as energy, the kinetic energy gained must be absorbable after clearing



$$t_{cc} = \max\{t_c: \delta(t) \text{ bounded } \forall t > t_c\}$$

## WHY IT MATTERS FOR A LEARNED MODEL

It tells you what the answer depends on: inertia, pre-fault loading, the post-fault coupling, and the clearing time. Those are the features a predictor must see.

It also gives a sanity check with no network in it: if your predictor disagrees with the equal-area verdict on a single-machine case, it is wrong.

## A TEST YOU CAN COMPUTE BY HAND

The one-machine-infinite-bus case has a closed-form answer. Notebook 01 keeps the habit for two machines: you sweep clearing times by simulation and check one label of the dataset before anything is learned.

# From a Verdict to a Number

## STABLE OR UNSTABLE IS NOT ENOUGH TO OPERATE ON

$$V(\delta, \omega) = \frac{1}{2} \sum_i M_i \omega_i^2 + V_{\text{pot}}(\delta)$$

$V$  transient energy,  $M_i$  inertia coefficient,  $\omega_i$  speed,  $V_{\text{pot}}$  the potential term

- a binary verdict tells an operator nothing about how close they are
- the energy formulation gives a margin instead
- positive means stable, and how positive says how much room is left

$$\eta = \frac{V_{\text{cr}} - V_{\text{cl}}}{V_{\text{cl}} - V_{\text{s}}}$$

$V_{\text{cr}}$  critical energy,  $V_{\text{cl}}$  energy at clearing,  $V_{\text{s}}$  at the stable equilibrium

### PREDICT THE MARGIN, NOT THE LABEL

|                  |                                       |
|------------------|---------------------------------------|
| <b>label</b>     | stable or unstable, no gradient       |
| <b>margin</b>    | how much room is left                 |
| <b>CCT</b>       | how much time is left                 |
| <b>all three</b> | derivable from a predicted trajectory |

### WHY PREDICT THE TRAJECTORY AT ALL

A classifier gives a verdict you cannot check. A predicted trajectory can be tested against the physics it should satisfy, plotted, and disbelieved on inspection. That auditability is most of the argument for the physics-informed route.

# What a Purely Data-Driven Predictor Does

## AND IT WORKS, UNTIL IT DOES NOT

$$p = g_{\theta}(\mathbf{x}_0, c) \in [0,1]$$

$g_{\theta}$  the learned model,  $\mathbf{x}_0$  pre-fault state,  $c$  the contingency,  $p$  probability of stability

- simulate many cases, label them, train a classifier on features of the pre-fault state
- it performs very well, because the test cases came from the training distribution
- nothing in the model knows a swing equation
- it has learned a correlation, and correlations do not survive a topology it has not seen

## WHAT IT GETS RIGHT

|                        |                              |
|------------------------|------------------------------|
| <b>speed</b>           | a forward pass, milliseconds |
| <b>in-distribution</b> | genuinely high               |
| <b>simplicity</b>      | no solver in the loop        |
| <b>as a filter</b>     | a defensible first pass      |

## THE FAILURE IS SILENT

An unstable case predicted stable produces no error message, no large residual and no warning — just a number an operator believes. Compare the topology failure in L12.1: the dangerous faults are the ones whose diagnostics look healthy.

# The Case You Must Get Right Is the One You Lack

## TRAINING SETS ARE MADE OF THINGS THAT HAPPENED

$$p_{\text{train}}(c) \neq p_{\text{test}}(c)$$

$c$  the contingency,  $p_{\text{train}}, p_{\text{test}}$  the two distributions over it

- grids are operated to be secure, so recorded history is overwhelmingly stable operation
- the cases you must get right are the ones you have least of
- **you can simulate them, and you should** — but then the training distribution is your simulator's
- a physics residual holds at states you never sampled, which is the point

## THE ONE-SENTENCE ARGUMENT

Data tells you what has happened; physics tells you what can. A predictor built only from the first is confident exactly where it is least entitled to be.

## WHERE THE SHIFT COMES FROM

|                        |                                     |
|------------------------|-------------------------------------|
| <b>new topology</b>    | an outage configuration not sampled |
| <b>operating point</b> | more wind, less inertia             |
| <b>rare fault</b>      | the one that matters, by definition |
| <b>future grid</b>     | converter-dominated, unlike history |

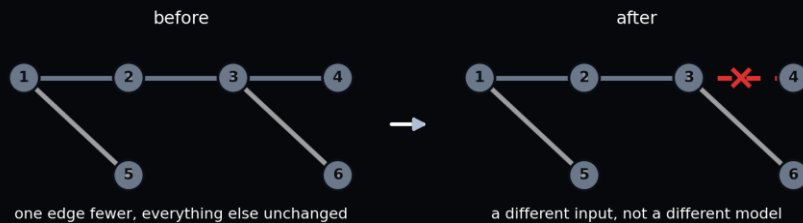
# A Contingency Is a Change to the Graph

## NOT A NEW PROBLEM — A NEW INPUT

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \quad \mathcal{G}_c = (\mathcal{V}, \mathcal{E} - \{e_c\})$$

$\mathcal{V}$  buses,  $\mathcal{E}$  branches,  $e_c$  the branch the contingency removes

- removing a branch is an edge deletion
- if the model takes the graph as an argument, the post-fault network is a different input
- a model that flattens the state into a fixed vector cannot do this



## THE SAME MODEL, A DIFFERENT GRAPH



## WHAT MUST STILL BE RECOMPUTED

$\mathbf{Y}$  changes with the topology, so the residual is rebuilt per contingency. Cheap and exact. What you avoid is retraining, not recomputing.

# Message Passing, and the Property That Matters

## THE RECAP, AND THEN THE POINT

$$h_{i^{(l+1)}} = \sigma \left( W_1 h_{i^{(l)}} + \sum_{j \in \mathcal{N}(i)} W_2 h_{j^{(l)}} \right)$$

$h_i^{(l)}$  node features,  $\mathcal{N}(i)$  the neighbours of  $i$ ,  $W_1, W_2$  shared weights,  $\sigma$  nonlinearity

- each node updates from its own features and an aggregate over its neighbours
- **the property that is appropriate here is equivariance** — permute the buses, the prediction permutes with them
- weight sharing follows from the same structure
- the learned update is local and applied everywhere, so it transfers between grids

$$g_\theta(P\mathbf{x}, P\mathcal{G}) = P g_\theta(\mathbf{x}, \mathcal{G})$$

$P$  a permutation of the buses,  $g_\theta$  the learned model

## THE MAPPING

|                      |                                      |
|----------------------|--------------------------------------|
| <b>node</b>          | a bus, or a machine                  |
| <b>node features</b> | $\delta, \omega, P_m, H$ at that bus |
| <b>edge</b>          | a line, present or tripped           |
| <b>edge features</b> | conductance and susceptance          |
| <b>layers</b>        | how far influence travels per step   |

## DEPTH IS A PHYSICAL CHOICE

Too few layers and a machine cannot feel a fault several buses away in one forward pass. Too many and every node sees the whole graph, the locality is gone, and training gets harder. Sweep it, and report the sweep.

# Physics-Informed Graph Network

## THREE TERMS, AND WHY EACH IS NECESSARY

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{ph}}\mathcal{L}_{\text{swing}} + \lambda_{\text{ic}}\mathcal{L}_{\text{ic}}$$

$\mathcal{L}_{\text{data}}$  the fit,  $\mathcal{L}_{\text{swing}}$  the physics,  $\mathcal{L}_{\text{ic}}$  the initial condition,  $\lambda$  their weights

- the data term fits simulated trajectories where you have them
- the physics term demands the prediction satisfy the swing equation
- the data term alone gives the data-driven predictor, better structured
- the residual is what survives distribution shift

### WHAT EACH TERM BUYS

|                          |                                       |
|--------------------------|---------------------------------------|
| <b>data</b>              | speed, and the shape of typical cases |
| <b>physics</b>           | behaviour where data runs out         |
| <b>initial condition</b> | built in, not a loss term             |
| <b>the weights</b>       | your statement of relative trust      |
| <b>together</b>          | auditable, and fast                   |

### THE WEIGHTS ARE THE EXPERIMENT

$\lambda_{\text{ph}} = 0$  is the data-driven baseline. Large  $\lambda_{\text{ph}}$  is a slow solver. The interesting behaviour is in between, and the exercise asks you to find it by sweeping — the L9.1 habit, for the third time in this course.

# Writing the Residual on a Graph

THE SWING EQUATION, EVALUATED WHERE YOU LIKE

$$\mathcal{L}_{\text{swing}} = \frac{1}{NT} \sum_i \sum_k \left\| \frac{2H_i}{\omega_s} \ddot{\delta}_i + D_i \dot{\delta}_i - P_{m,i} + P_{e,i} \right\|^2$$

$N$  machines,  $T$  collocation times,  $H_i$  inertia,  $D_i$  damping,  $P_{e,i} - P_{m,i}$  the power imbalance

- one residual per machine per collocation time
- time derivatives come from autograd, exactly as in L7
- **the electrical power term is where the graph enters** — it sums over post-fault neighbours
- and collocation points should not all sit in the easy part of the window

## TWO WAYS TO GET THE DERIVATIVES

|                        |                                                           |
|------------------------|-----------------------------------------------------------|
| <b>continuous</b>      | network of $t$ ; autograd gives $\delta$ , $\dot{\delta}$ |
| <b>rollout</b>         | discrete step; residual on the update                     |
| <b>continuous, pro</b> | exact derivatives, free collocation                       |
| <b>rollout, pro</b>    | matches how it will be deployed                           |

## THE SECOND DERIVATIVE COSTS

$\delta$  means differentiating twice through the network, which costs and can be noisy early in training. Writing the system in first-order form — as in L12.1 — avoids it entirely and is usually the better choice.

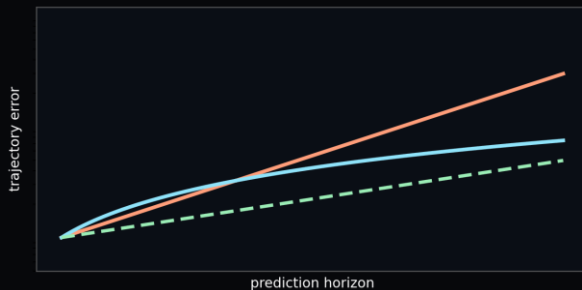
# One Shot, or Step by Step

## AND THE ERROR BEHAVES DIFFERENTLY IN EACH

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_k + \Delta t g_{\theta}(\hat{\mathbf{x}}_k, \mathcal{G}_c)$$

$\hat{\mathbf{x}}_k$  the predicted state,  $\Delta t$  the step,  $\mathcal{G}_c$  the post-fault graph

- a rollout applies a learned update repeatedly, feeding its own output back
- **it matches the physics and it compounds its own errors** — L6.2's problem, on a trajectory
- a one-shot model maps the initial state and the contingency straight to the answer



### PICK BY HORIZON

|                        |                                |
|------------------------|--------------------------------|
| <b>short horizon</b>   | either works                   |
| <b>long horizon</b>    | rollout drifts without physics |
| <b>margin only</b>     | one-shot is enough             |
| <b>full trajectory</b> | rollout, with the residual on  |

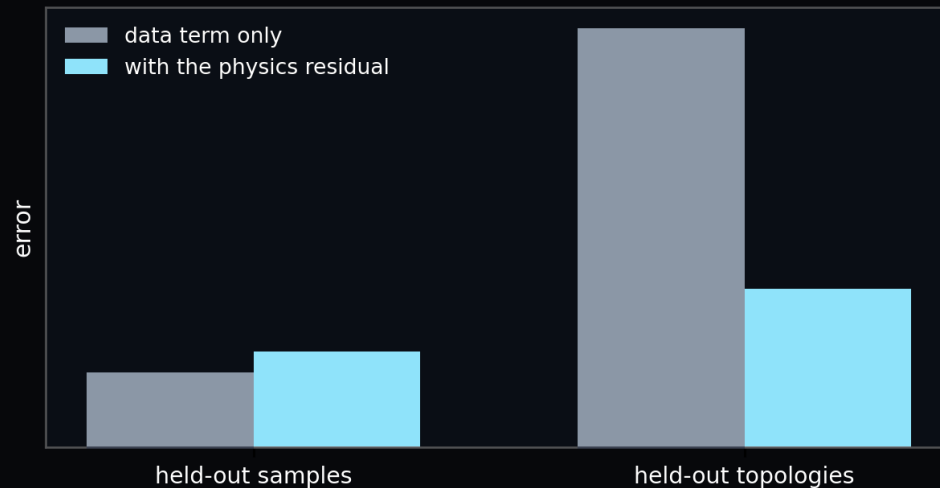
### TEST TO FAILURE, NOT TO SUCCESS

Do not report the horizon at which your model works. Report the horizon at which it stops working, for each variant. The second number is informative and the first is a choice of where to stop plotting.

# Held-Out Topologies, Not Held-Out Samples

## SPLIT THE DATA THE WAY THE CLAIM REQUIRES

The usual random split tells you nothing here. Training and test cases drawn from the same set of topologies share the thing you claim to generalise over, and a high score means only that the model interpolates within it.



## THE FOUR-WAY COMPARISON

|                      |                            |
|----------------------|----------------------------|
| <b>MLP, seen</b>     | the flattering number      |
| <b>MLP, unseen</b>   | usually collapses          |
| <b>GNN, unseen</b>   | structure helps            |
| <b>PIGNN, unseen</b> | the claim, tested          |
| <b>simulation</b>    | the truth, for all of them |

## AN HONEST NEGATIVE IS STILL A RESULT

Ex\_12.2 builds the first three rows; no notebook builds the PIGNN. If one does not beat the GNN on held-out topologies, say so and find out why. A reported negative is worth more than a tuned positive.

# What to Build In Rather Than Penalise

## THE THIRD TIME THIS IDEA APPEARS IN THE COURSE

$$\omega_i = \omega_s + \Delta\omega_{\max} \tanh(\hat{\omega}_i)$$

$\hat{\omega}_i$  the raw network output,  $\omega_s$  synchronous speed,  $\Delta\omega_{\max}$  the bound imposed

The initial condition is known exactly — it is the estimate you start from — so impose it by construction rather than by a loss term, exactly as L7.2 imposed the initial condition on the heat equation.

Speed deviations have a physical range; a machine at fifty per cent over synchronous speed has left the model's validity long before it left the optimiser's search space. Bounding the output keeps predictions interpretable when they are wrong.

Angle differences are the one to leave free. Bounding them would prevent the model from ever predicting loss of synchronism — which is the event you built the whole thing to see.

### HARD, SOFT, FREE

|                          |                                  |
|--------------------------|----------------------------------|
| <b>initial condition</b> | hard — build into the trial form |
| <b>speed range</b>       | hard — bounded activation        |
| <b>swing equation</b>    | soft — the residual, weighted    |
| <b>angle separation</b>  | free — never bound it            |

### THE MISTAKE THIS PREVENTS

Constraining the quantity whose excursion defines the failure guarantees a model that never predicts failure. It is the same error as imposing thermal limits in L12.1: a comfortable picture of a system in trouble.

# How This Fails

## FOUR, AND THE FIRST TWO ARE SILENT

- **confident off-distribution** — an unfamiliar case returns a smooth, plausible, wrong trajectory, with no warning
- **residual satisfied, answer wrong** — the model omits governors and protection, so a wrong answer can satisfy it
- **rollout drift** — each step is accurate and a hundred are not, visible only if you test past your horizon
- **the initial condition was wrong** — everything starts from L12.1's estimate, and an error there becomes a verdict here

### THE ONE DIAGNOSTIC YOU HAVE

Evaluate the physics residual at inference, not only during training, and treat a large value as a refusal. A physics-informed model can police itself this way. A classifier cannot.

### WHY THE FIRST TWO ARE SILENT

Neither announces itself. The output is smooth and plausible, so nothing in the prediction tells you which regime you are in.

### AND THEY FAIL TOGETHER

No amount of care in this lecture recovers an estimation error made in L12.1.

# Why Any of This Should Run Locally

## TOO LATE IS NOT A PREDICTION

Transient stability plays out in the first seconds after a fault. A verdict that reaches a control room after the machines have already separated is a post-mortem, however accurate it is.

Central assessment involves getting measurements to a control centre, computing, and getting a decision back. Each leg costs time, and the total is what must fit inside the window the physics allows — not the compute alone.

Local inference removes the two communication legs. That is the entire argument for the edge here: not that the hardware is fast, but that the distance is short.

It also changes what can fail. A local predictor keeps working when the link to the control centre does not, which is exactly the condition under which grids get interesting.

## WHAT LOCAL BUYS

|                     |                                     |
|---------------------|-------------------------------------|
| <b>latency</b>      | two comms legs removed              |
| <b>availability</b> | survives loss of the link           |
| <b>bandwidth</b>    | send verdicts, not raw streams      |
| <b>privacy</b>      | measurements stay in the substation |

## AND WHAT IT COSTS

A model in a substation is a model somebody must update, version and validate in the field, on hardware in an electrically hostile place with a ten-year service life. The engineering problem moves; it does not disappear.

# Where the Milliseconds Actually Go

## ADD THEM UP BEFORE CHOOSING ANY HARDWARE

$$t_{\text{tot}} = t_{\text{meas}} + t_{\text{comm}} + t_{\text{inf}} + t_{\text{act}}$$

Four terms. Measurement — a PMU reporting at 50 Hz cannot tell you anything sooner than its reporting interval, whatever follows it. Communication. Inference. Actuation, which is a physical device closing or opening.

For a small network the inference term is the smallest of the four by a wide margin, and it is the only one people benchmark. That is the trap: optimising the term that was never binding.

The exercise asks you to measure all four on your own setup and present them as a budget. Only then does the question of hardware become answerable, because only then do you know how much time you were trying to buy.

measure

comms

infer

act

*indicative proportions only — inference is the thin one*

## THE CONSTRAINT

$$t_{\text{tot}} \ll t_{\text{cc}}$$

The total must be comfortably inside the critical clearing time. That is the link between the physics and the engineering.

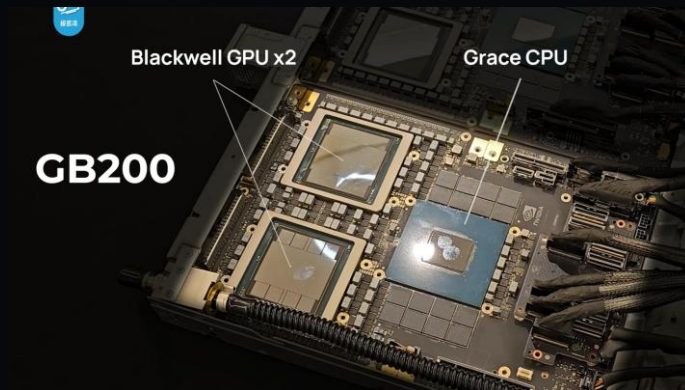
## THE L11.1 PARALLEL

Blind distance was  $d = v/f$ : how far the car travelled between seeing and acting. The same quantity for a grid, measured rather than assumed.

# Jetson Thor, and What It Is Actually For

## THE BOARD IN THE LAB

- **based on GB200** — the same GPU plus CPU Blackwell architecture
- **for physical autonomous systems** — robots, self-driving cars
- **local edge AI for grid control** — power flow for hundreds of nodes



## THE JETSON THREAD, CLOSED

|                |                                     |
|----------------|-------------------------------------|
| compute        | 2 070 TFLOPS sparse FP4, Blackwell  |
| CPU            | 14-core Arm Neoverse V3AE           |
| memory         | 128 GB LPDDR5X, 273 GB/s            |
| power          | 40 to 130 W, configurable           |
| what it is for | many contingencies at once, on site |

## NOT WHAT YOU WOULD DEPLOY

A substation device is a hardened IED with a decade of service life and IEC 61850 interfaces — not a development kit drawing over a hundred watts. Thor is what you learn on; say which is which.

# Getting a Trained Model onto the Board

## AND WHAT CHANGES WHEN YOU DO

$$r = \frac{1}{t_{\text{inf}}} \geq N_c f_{\text{screen}}$$

$r$  throughput,  $t_{\text{inf}}$  inference time,  $N_c$  contingencies to screen,  $f_{\text{screen}}$  the refresh rate

Training happens on a workstation; inference happens on the board. Between them sits an export step and an optimisation step, and both can change numerical behaviour — reduced precision first among them.

Screening is naturally batched: many contingencies, the same weights, independent graphs. Throughput matters as much as single-case latency. The requirement is to screen the whole list within the interval you refresh it.

Check the physics after optimising, not just the accuracy. If quantisation degrades the residual on cases the model handled before, you have traded correctness for speed you may not have needed — and the residual will tell you, where a test-set score might not.

### THE STEPS

|                  |                                   |
|------------------|-----------------------------------|
| <b>export</b>    | out of the training framework     |
| <b>optimise</b>  | for the target device             |
| <b>precision</b> | reduced, then re-check            |
| <b>batch</b>     | many contingencies at once        |
| <b>measure</b>   | latency and throughput, on device |

### MEASURE ON THE DEVICE, WARMED UP

First-call timings include compilation and allocation. Discard them, run many iterations, and report a distribution rather than a best case — the L11 lap timing discipline.

# What Belongs Local, and What Does Not

## NOT A REPLACEMENT FOR THE CONTROL CENTRE

A local predictor sees local measurements and a local model. It cannot know the state of the whole system, it cannot be retrained in the substation, and it should not be making decisions that require a system-wide view.

What it can do is act fast on the cases where speed is the binding constraint, and flag rather than decide when its own residual says it is outside what it understands. Refusal is a legitimate output and should be designed in from the start.

The control centre keeps the full model, the training, the validation and the authority. This division — fast and local, careful and central — is older than machine learning in this industry, and the new component fits into it rather than replacing it.

### LOCAL

|                       |                               |
|-----------------------|-------------------------------|
| <b>inference</b>      | on the current window         |
| <b>residual check</b> | am I outside what I know      |
| <b>fast flags</b>     | when the budget is tight      |
| <b>buffering</b>      | keep data when the link drops |

### CENTRAL

|                    |                                 |
|--------------------|---------------------------------|
| <b>training</b>    | and every retraining            |
| <b>validation</b>  | against full simulation         |
| <b>system view</b> | coordinating across substations |

## Ex\_12.2 — What You Will Build

### SEVEN NOTEBOOKS, ALL OF THEM ON COLAB

You begin from the network and the estimated state of Ex\_12.1, generate a contingency set by simulation, and hold out whole topologies rather than random samples. That split is the experiment; everything else supports it.

Notebook 02 trains the data-driven baseline and reports its flattering in-distribution score. Notebook 03 adds the graph structure and holds a whole topology out — the comparison the method rests on.

Notebook 04 turns a regressor into a screen by choosing a threshold, and notebook 06 assembles the latency budget — then say whether it would have been fast enough.

#### WHERE EACH PART RUNS

Everything here is Colab. You report ratios that carry across hardware; L13 runs the same script on a real Thor.

#### THE NOTEBOOKS

|                          |                                   |
|--------------------------|-----------------------------------|
| <b>00 · 01 set-up</b>    | network, then contingencies       |
| <b>02 dense baseline</b> | in-distribution — has TODOs       |
| <b>03 graph network</b>  | topology as input — has TODOs     |
| <b>04 screening</b>      | the threshold, and the two errors |
| <b>05 compare</b>        | the table and the report          |
| <b>06 latency</b>        | throughput and the budget         |

#### THE QUESTION THAT CARRIES THE MARKS

Would you let this run unsupervised in a substation? The answer is no, and the marks are for saying what would have to be true first.

# The Orderings That Must Come Out Right

## THE RELATIONS, NOT THE NUMBERS

On seen topologies every method does well, including the MLP. That is the control, and a result that disagrees with it means something is wrong with the implementation rather than interesting about the physics.

On held-out topologies the MLP should degrade sharply and the GNN less; a PIGNN, not built here, least. The size of the gaps depends on your setup; the ordering should not.

On the board, inference should be a small fraction of the total budget. If it is not, either your model is far larger than the problem needs or your measurement includes warm-up.

Add a physics term and its residual should rise where accuracy falls. If it does not, the term is not constraining what you think it is — worth finding out in a notebook rather than in a substation.

## THE CHECKLIST

|                        |                             |
|------------------------|-----------------------------|
| <b>seen topologies</b> | all methods accurate        |
| <b>held-out</b>        | MLP worst, GNN better       |
| <b>relabelling</b>     | GNN exact, MLP destroyed    |
| <b>threshold</b>       | zero misses costs alarms    |
| <b>latency</b>         | inference is the thin slice |
| <b>hand label</b>      | agrees with the dataset     |

## ANY ONE OF THEM BACKWARDS

Six relations. A reversed ordering points at the implementation rather than at the physics, and that is the fastest debugging tool in the exercise.

# What This Method Is Not Ready To Do

## STATED PLAINLY, BECAUSE SOMEBODY WILL ASK

- **not a replacement for simulation** — it is a filter; simulation stays the answer of record
- **it inherits the model's limits** — no exciter, no governor, no protection, so the residual cannot see those failures
- **it carries no guarantee** — no covariance, no bound, no certificate; the residual is evidence, not proof
- **not tested at scale here** — a few dozen buses is a teaching case; the real argument lives at hundreds or thousands

### WHAT IT IS FOR, THEN

Screening. Answering the same question thousands of times more cheaply than integration can, so integration is spent where it matters. A screening tool that is wrong occasionally is useful; a tool of record that is wrong occasionally is not.

### THE QUESTION TO ASK ANYONE

Ask what exactly was verified, and over what set.

### AND THE STANDARD FOR YOUR REPORT

State the boundary of your claim first.

# Questions

## 1 – 2 · TRANSIENT STABILITY

### 1 Which stability question is asked here, and why is screening combinatorial?

Rotor-angle stability: after a fault, do the machines stay in step? Every operating state times every possible outage is a case, and each needs a simulation in time, so the cases outgrow the time available. A learned predictor screens them and leaves the simulation for the doubtful ones.

*Kundur ch. 2, 13; Ex\_12.2 notebooks 01, 04*

### 2 What does the equal area criterion say, and why is a margin worth more than a verdict?

Read as energy: the machine stays in step if the area that can slow it down after the fault is cleared is at least the area that sped it up during the fault. The difference is a margin, which tells an operator how close a case is; a verdict says only stable or not. *Kundur ch. 13; Ex\_12.2 notebook 01*

## 3 – 4 · GRAPH PREDICTOR

### 3 Why is a contingency a change to the graph, and what does a graph network gain from that?

A tripped line is an edge removed, not a new value in a feature vector. A graph network applies the same local rule on whatever graph it is given, so it can be tested on topologies it has never seen, which a dense network cannot represent at all. *Ex\_12.2 notebooks 02 – 03; L5.1*

### 4 Why hold out topologies and not samples, and what is the method not ready to do?

Held-out samples of topologies seen in training measure interpolation and flatter a model with no physics in it; the unstable cases are the rare ones, and the ones that must be right. Held-out topologies are the fair test. The predictor screens; it is not a tool of record. *Ex\_12.2 notebooks 03 – 05*

# What You Have Actually Been Learning

## ONE METHOD, FIVE FIELDS

|                             |                                            |
|-----------------------------|--------------------------------------------|
| <b>L8 heat</b>              | a plate, steady then transient             |
| <b>L9 flow</b>              | a cavity, a cylinder, then turbulence      |
| <b>L10 electrochemistry</b> | a cell, then a stack, then an optimisation |
| <b>L11 robotics</b>         | a policy, then constraints and energy      |
| <b>L12 the grid</b>         | a state, then a prediction                 |

### THE ARC, IN ONE SENTENCE

Part 1 taught you what a network is. Part 2 taught you to put a physical law where the labels would go.

## AND ONE STRUCTURE, THROUGHOUT

|                     |                                    |
|---------------------|------------------------------------|
| <b>a network</b>    | proposes a field, or a decision    |
| <b>a residual</b>   | judges it against the physics      |
| <b>a constraint</b> | built in, not penalised            |
| <b>an objective</b> | when you want more than a solution |

## AND WHAT YOU SHOULD BE SCEPTICAL OF

Every lecture in Part 2 named a case where a classical method is better. That list is the useful one.

## WHERE IT GOES NEXT

L13 takes five of these fields further, as mini-projects where the answer is not known in advance.

## One Line to Remember

*Data tells you what has happened. Physics tells you what can. The case you must get right is, by construction, the one you have no data for.*

### AND THREE THINGS THAT FOLLOW

Test on held-out topologies, not held-out samples. Any other split measures interpolation and calls it generalisation.

Let the model refuse. A physics residual evaluated at inference is a self-check no purely data-driven predictor has, and it is the difference between a demonstration and something deployable.

Measure the latency budget before choosing hardware. The term everyone benchmarks is rarely the one that binds.

### IF YOU DO ONE THING

Build the four-way table on held-out topologies. It is a day's work, it is the experiment the whole lecture argues for, and it will tell you whether the physics term earned its place in your setup rather than in a paper's.

A negative result there, honestly reported, is a better piece of engineering than a tuned positive.

# References and Further Reading

## POWER SYSTEM DYNAMICS

Kundur (1994), Power System Stability and Control — transient stability, the equal area criterion, machine models. Chapter 13 is the one for this lecture.

Machowski, Bialek & Bumby, Power System Dynamics — a modern treatment with more on control and protection.

Pai, Energy Function Analysis for Power System Stability — the energy-margin formulation used here.

## LEARNING, AND THE GRAPH

Raissi, Perdikaris & Karniadakis (2019), J. Comput. Phys. 378, 686–707 — the physics-informed formulation used all course.

The graph network material from Part 1 of this course — message passing, and why equivariance matters.

The physics-informed GNN literature for power systems is active and young. Reproduce before you cite, and be sceptical of any accuracy figure reported without a held-out-topology split.

## DATA AND TOOLS

|                        |                                   |
|------------------------|-----------------------------------|
| <b>Energinet</b>       | Danish injections and exchanges   |
| <b>Fingrid, SvK</b>    | Nordic frequency, real events     |
| <b>ENTSO-E</b>         | pan-European operational data     |
| <b>PyPSA-Eur</b>       | open European network topology    |
| <b>IEEE test cases</b> | small and exact, for verification |
| <b>NVIDIA Jetson</b>   | the board in the lab              |

*Everything in the exercise runs on torch, numpy and matplotlib, plus a graph library for notebooks 02 onward and the vendor toolchain for notebook 04.*

END OF THE COURSE

# Applied PINN for Energy

From a Poisson equation on a unit square to a stability predictor on a national grid, with the same four moves throughout. What you take away is the habit, not the architecture.

## TO FINISH

|                    |                                     |
|--------------------|-------------------------------------|
| <b>Ex_12.2</b>     | the held-out table, and the latency |
| <b>the report</b>  | state the boundary of your claim    |
| <b>MiniProject</b> | turbulence closure, if you took L13 |
| <b>afterwards</b>  | apply it outside this course        |

*The most useful thing you can do next is take a problem you were not set, and ask what physics you already trust about it.*